Lecture 13: Input Resistance and the Δ–Y Transform

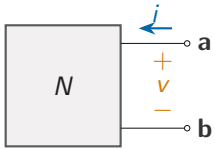
EE201 – Circuit Theory I

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Department of Electrical and Electronics Engineering
Middle East Technical University

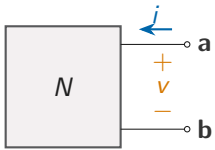
current voltage, node voltage e Δ Y

Review: equivalent one-ports



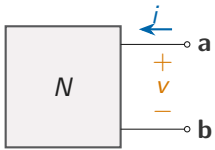
- One-port: a port i, v

Review: equivalent one-ports



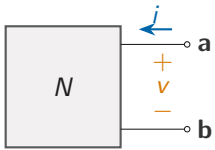
- One-port: a port i, v
- Equivalent: the same i–v characteristic

Review: equivalent one-ports



- One-port: a port i, v
- Equivalent: the same i–v characteristic
- Series, parallel
- Source transformation

Review: equivalent one-ports



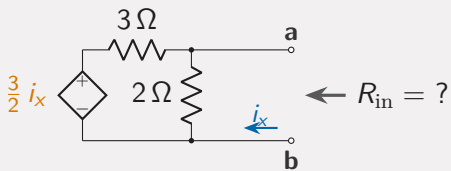
- One-port: a port i, v
- Equivalent: the same i–v characteristic
- Series, parallel
- Source transformation

Node and mesh analysis: the original circuit

Equivalents: a simpler circuit

Motivation

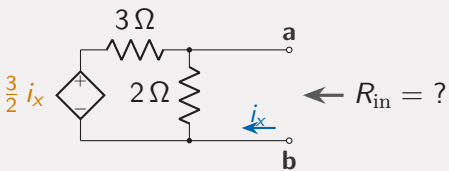
Dependent sources



Series, parallel: not enough

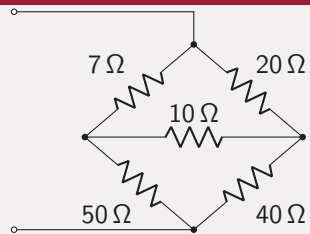
Motivation

Dependent sources



Series, parallel: not enough

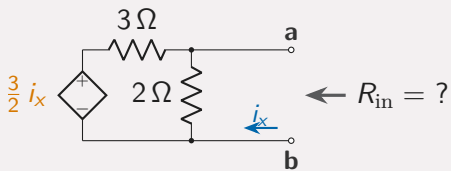
A bridge



No series, no parallel

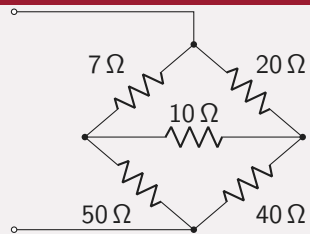
Motivation

Dependent sources



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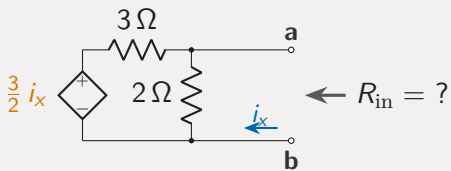


No series, no parallel

One equivalent resistor in both?

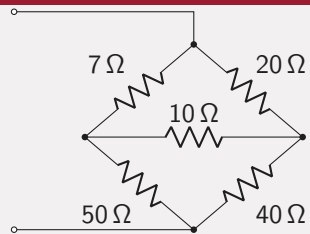
Motivation

Dependent sources



Series, parallel: not enough

A bridge



No series, no parallel

One equivalent resistor in both? Yes

Roadmap



Input resistance

1

the idea: a test source

with a dependent source



Δ to Y

2

the same resistance between terminals



Y to Δ

3

the inverse relations

equal resistors

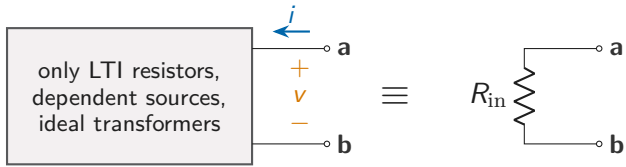


An example

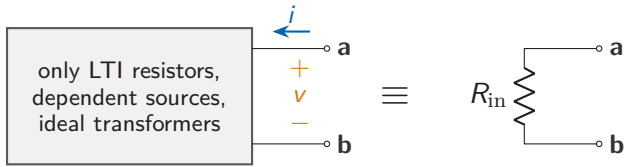
4

a bridge

One-ports without independent sources

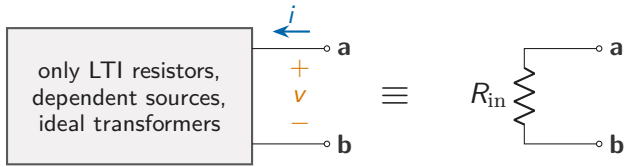


One-ports without independent sources



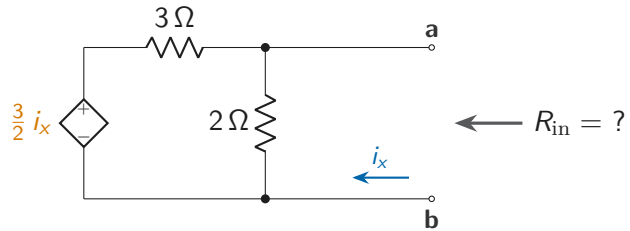
- Controls of the dependent sources: inside the one-port

One-ports without independent sources

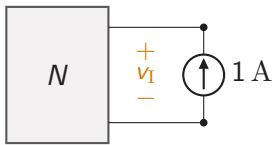


- Controls of the dependent sources: inside the one-port
- $v = R_{\text{in}}\, i$: R_{in} is the input resistance

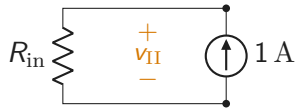
Example 13.1: an input resistance



The idea: a test source

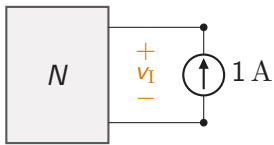


(I)

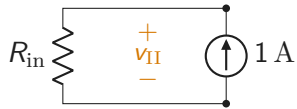


(II)

The idea: a test source



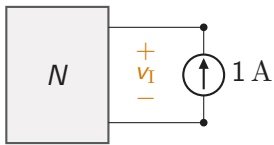
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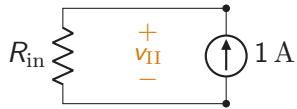
(II)

$$N \equiv R_{\mathrm{in}} \;\Rightarrow\; v_{\mathrm{I}} = v_{\mathrm{II}} = R_{\mathrm{in}} \cdot 1\,\mathrm{A}$$

The idea: a test source



(I)

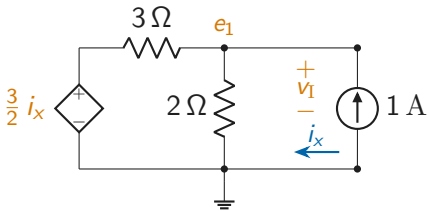


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$$N \equiv R_{\mathrm{in}} \;\Rightarrow\; v_{\mathrm{I}} = v_{\mathrm{II}} = R_{\mathrm{in}} \cdot 1\,\mathrm{A}$$

$$R_{\mathrm{in}} = \frac{v_{\mathrm{I}}}{1\,\mathrm{A}}\text{: solve circuit (I)}$$

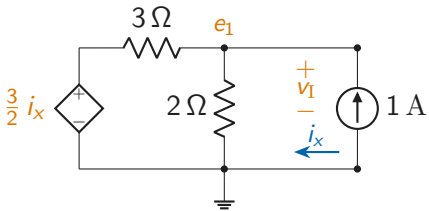
Example 13.1: the solution



Node 1:

$$\frac{e_1 - \frac{3}{2}\, i_x}{3} + \frac{e_1}{2} - 1 = 0$$

Example 13.1: the solution

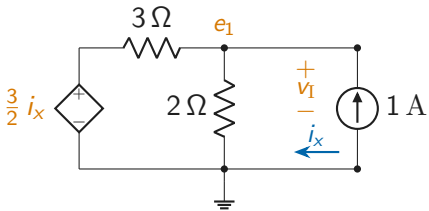


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Example 13.1: the solution



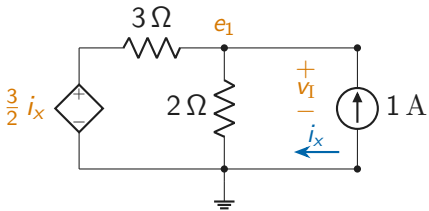
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$i_x = -1\,\text{A}$

Example 13.1: the solution



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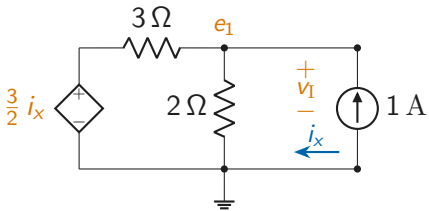
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$$\frac{e_1 + \frac{3}{2}}{3} + \frac{e_1}{2} - 1 = 0 \;\Rightarrow\; e_1 = \tfrac{3}{5}\,\text{V}$$

Example 13.1: the solution



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$$\frac{e_1 + \frac{3}{2}}{3} + \frac{e_1}{2} - 1 = 0 \;\Rightarrow\; e_1 = \tfrac{3}{5}\,\mathrm{V}$$

$$R_\mathrm{in} = \frac{v_\mathrm{I}}{1\,\mathrm{A}} = \frac{3}{5}\,\Omega$$

Roadmap



Input resistance

✓

the idea: a test source ✓

with a dependent source ✓



Δ to Y

2

the same resistance between terminals



Y to Δ

3

the inverse relations

equal resistors

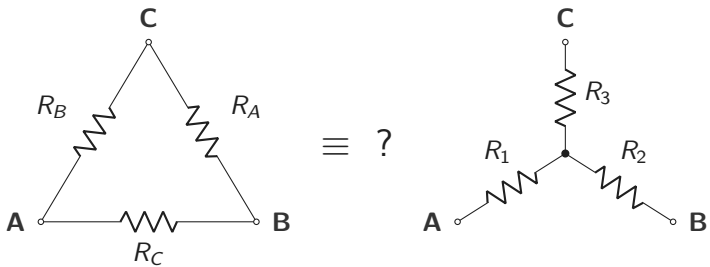


An example

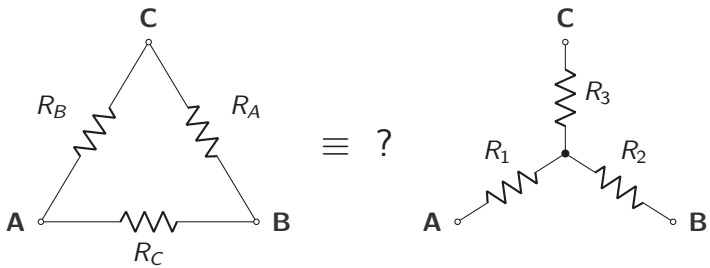
4

a bridge

Three-terminal circuits: Δ and Y

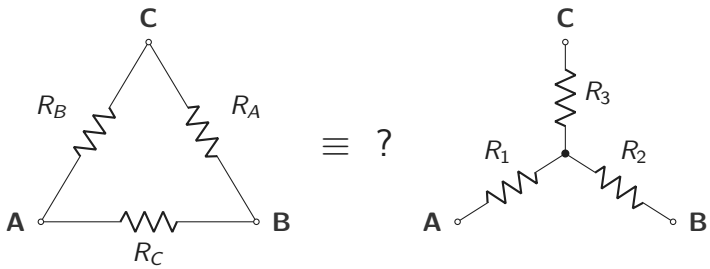


Three-terminal circuits: Δ and Y



Given R_A, R_B, R_C: find R_1, R_2, R_3

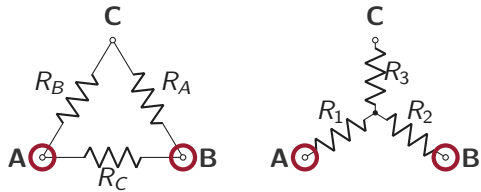
Three-terminal circuits: Δ and Y



Given R_A, R_B, R_C: find R_1, R_2, R_3

The same resistance between every two terminals

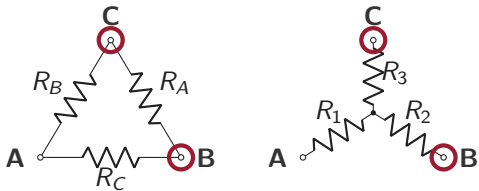
Δ to Y: the resistance between two terminals



Δ Y

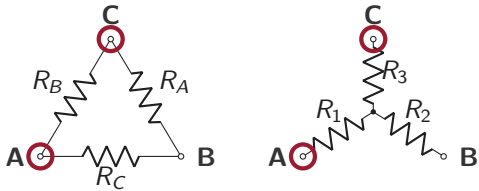
A–B: $$\frac{R_C(R_A + R_B)}{R_A + R_B + R_C} = R_1 + R_2 \quad (1)$$

Δ to Y: the resistance between two terminals



	Δ		Y	
A–B:	$\dfrac{R_C(R_A+R_B)}{R_A+R_B+R_C}$	$=$	R_1+R_2	(1)
B–C:	$\dfrac{R_A(R_B+R_C)}{R_A+R_B+R_C}$	$=$	R_2+R_3	(2)

Δ to Y: the resistance between two terminals



	Δ		Y	
A–B:	$\dfrac{R_C(R_A+R_B)}{R_A+R_B+R_C}$	$=$	R_1+R_2	(1)
B–C:	$\dfrac{R_A(R_B+R_C)}{R_A+R_B+R_C}$	$=$	R_2+R_3	(2)
C–A:	$\dfrac{R_B(R_C+R_A)}{R_A+R_B+R_C}$	$=$	R_3+R_1	(3)

Δ to Y: the result

$$\frac{1}{2}\left[(1)+(2)+(3)\right]: \quad R_1+R_2+R_3=\frac{R_AR_B+R_BR_C+R_CR_A}{R_A+R_B+R_C} \tag{4}$$

Δ to Y: the result

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$$R_1=\frac{R_BR_C}{R_A+R_B+R_C} \quad (4)-(2)$$

Δ to Y: the result

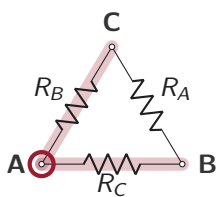
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$$R_1=\frac{R_BR_C}{R_A+R_B+R_C} \quad (4)-(2) \qquad R_2=\frac{R_CR_A}{R_A+R_B+R_C} \qquad R_3=\frac{R_AR_B}{R_A+R_B+R_C}$$

Δ to Y: the result

$$\tfrac{1}{2}\big[(1)+(2)+(3)\big]: \quad R_1+R_2+R_3=\frac{R_AR_B+R_BR_C+R_CR_A}{R_A+R_B+R_C} \qquad (4)$$

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Y resistor at a terminal = product of the two Δ resistors there / sum of the Δ resistors

Roadmap



Input resistance

✓

the idea: a test source ✓

with a dependent source ✓



Δ to Y

✓

the same resistance between terminals ✓



Y to Δ

3

the inverse relations

equal resistors



An example

4

a bridge

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

$$R_1R_2 + R_2R_3 + R_3R_1 = \frac{R_AR_BR_C\,(R_A + R_B + R_C)}{(R_A + R_B + R_C)^2} = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

$$R_1R_2 + R_2R_3 + R_3R_1 = \frac{R_AR_BR_C\,(R_A + R_B + R_C)}{(R_A + R_B + R_C)^2} = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

$$R_AR_1 = R_BR_2 = R_CR_3 = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

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$$R_AR_1 = R_BR_2 = R_CR_3 = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

$$R_A = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}, \qquad R_B = \frac{\cdots}{R_2}, \qquad R_C = \frac{\cdots}{R_3}$$

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

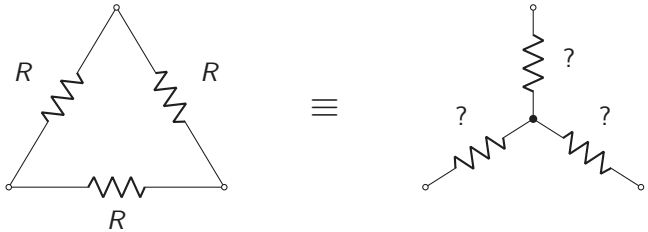
$$R_1R_2 + R_2R_3 + R_3R_1 = \frac{R_AR_BR_C\,(R_A + R_B + R_C)}{(R_A + R_B + R_C)^2} = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

$$R_AR_1 = R_BR_2 = R_CR_3 = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

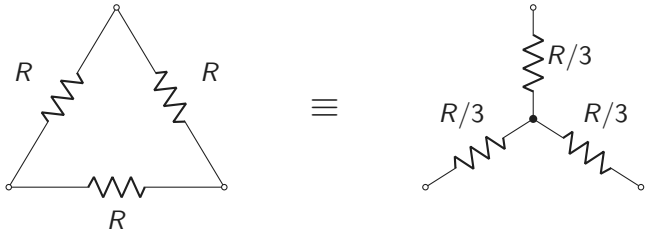
$$R_A = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}, \qquad R_B = \frac{\cdots}{R_2}, \qquad R_C = \frac{\cdots}{R_3}$$

$$\Delta \text{ resistor} = \frac{\text{sum of the products of the Y resistors}}{\text{the Y resistor opposite}}$$

Equal resistors

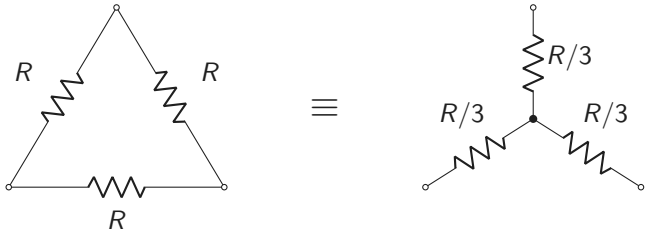


Equal resistors



$$\frac{R \cdot R}{3R} = \frac{R}{3}$$

Equal resistors



$$\frac{R \cdot R}{3R} = \frac{R}{3}$$

Y with $R \equiv \Delta$ with $3R$

Roadmap



Input resistance ✓

the idea: a test source ✓

with a dependent source ✓



Δ to Y ✓

the same resistance between terminals ✓



Y to Δ ✓

the inverse relations ✓

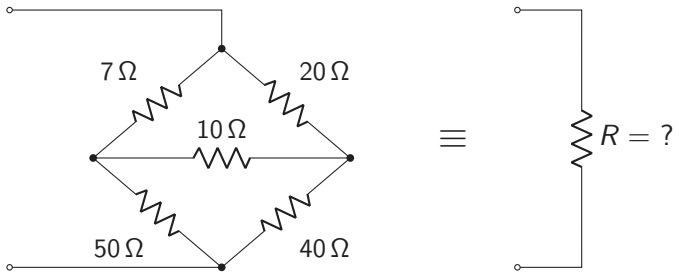
equal resistors ✓



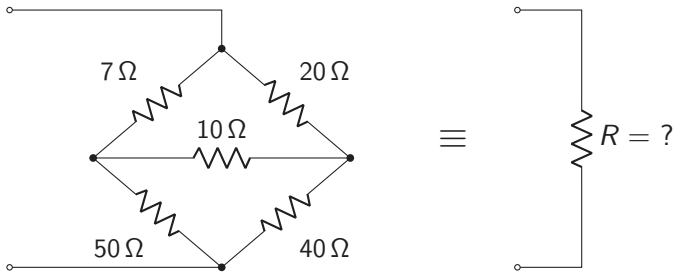
An example 4

a bridge

Example 13.2: a bridge

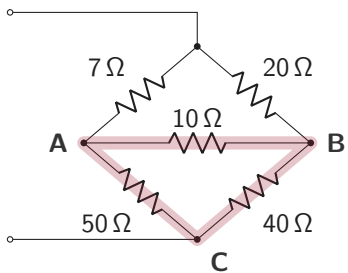


Example 13.2: a bridge



No two resistors in series or in parallel

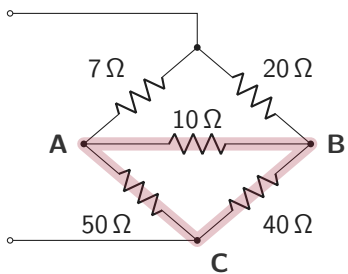
Example 13.2: Δ to Y



Δ A–B–C:

$R_C = 10\,\Omega, \quad R_B = 50\,\Omega, \quad R_A = 40\,\Omega$

Example 13.2: Δ to Y

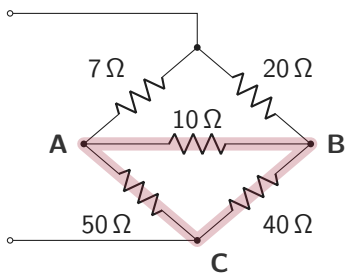


Δ A–B–C:

$R_C = 10\,\Omega, \quad R_B = 50\,\Omega, \quad R_A = 40\,\Omega$

$R_A + R_B + R_C = 100\,\Omega$

Example 13.2: Δ to Y



Δ A–B–C:

$R_C = 10\,\Omega, \quad R_B = 50\,\Omega, \quad R_A = 40\,\Omega$

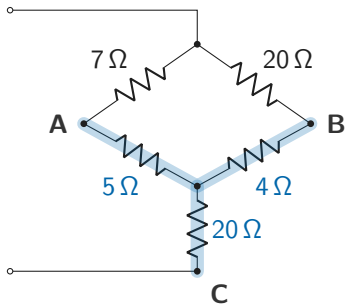
$R_A + R_B + R_C = 100\,\Omega$

$$R_1 = \frac{50 \cdot 10}{100} = 5\,\Omega$$

$$R_2 = \frac{10 \cdot 40}{100} = 4\,\Omega$$

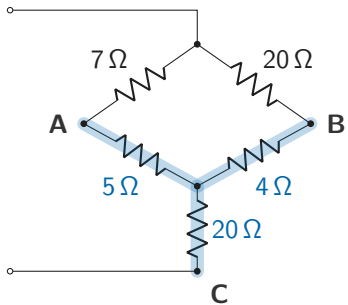
$$R_3 = \frac{40 \cdot 50}{100} = 20\,\Omega$$

Example 13.2: the new circuit



Series: $7 + 5 = 12\,\Omega, \quad 20 + 4 = 24\,\Omega$

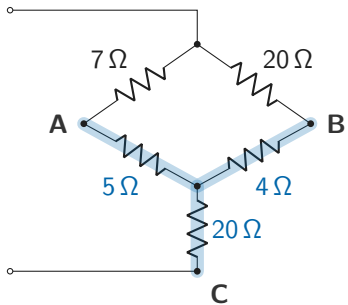
Example 13.2: the new circuit



Series: $7 + 5 = 12\,\Omega, \quad 20 + 4 = 24\,\Omega$

Parallel: $12 \parallel 24 = 8\,\Omega$

Example 13.2: the new circuit



Series: $7 + 5 = 12\,\Omega, \quad 20 + 4 = 24\,\Omega$

Parallel: $12 \parallel 24 = 8\,\Omega$

Series: $8 + 20$

$$R = 28\,\Omega$$

Summary

In the circuit	What to do
resistors, dependent sources	$R_{\text{in}} = v/1\,\text{A}$ with a $1\,\text{A}$ test source
a Δ	Y: $R_1 = R_B R_C/(R_A + R_B + R_C)$, ...
a Y	Δ: $R_A = (R_1R_2 + R_2R_3 + R_3R_1)/R_1$, ...
equal resistors	Δ with R $\equiv$ Y with $R/3$